

IMO 2026 — Problem 6

Solution by Claude Fable 5

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Source: `results/claude-fable-5/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

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Approaches tried

- **Simulation / pattern finding** (`code/verify.py`): for many seeds a_1 the set $S = \{a_n\}$ is exactly $[a_1, \infty) \cap \bigcup_Q d_Q \mathbb{Z}$ for a finite family of squarefree moduli d_Q (e.g. $a_1 = 15$: moduli 6, 10, 15; $a_1 = 385$: moduli 30, 66, 14, 418, 231, 399, 385), and $a_{n+T} = a_n + L$ holds from $n = 1$. This suggested the exact structural characterization.
- **Self-consistency + blocker (hypergraph transversal) reduction**: proved $S = \{m \geq a_1 : \gcd(m, s) > 1 \ \forall s \in S\}$ and hence $S = [a_1, \infty) \cap \bigcup_{Q \in \mathcal{M}} d_Q \mathbb{Z}$ where $\mathcal{M}$ is the set of minimal transversals of the family of prime-divisor sets of elements of S . Reduced the problem to: $\mathcal{M}$ is finite.

- **Dead end — pure set-system arguments** (approaches/abstract-clutter-dead-end.md): tried to prove finiteness of $\mathcal{M}$ from “self-blocking clutter” axioms alone (pairwise intersecting antichain in which every finite transversal contains a member). Found an explicit **infinite** self-blocking clutter of finite sets $(U_k(u) = \{u_1, \dots, u_k, x_{k+1}, y_{k+1}\}, u_i \in \{x_i, y_i\})$, so the greedy (“smallest next term”) property must be used quantitatively, not just the fixed-point property of S .
- **Successful approach** (approaches/main-blocker-descent.md): from greediness, every non-element of S has a *smaller* coprime witness in S . Applied to $m = \pi(Q \setminus \{t\})^e$ this yields a “cheap witness” member Q' with $Q' \cap Q = \{t\}$ and $d_{Q'} < a_1 d_Q / t$. Iterating at a fixed prime $t = q > a_1$ gives a geometric descent $d \mapsto (a_1/q)d$ of member products, an impossibility; hence all primes occurring in members are $\leq a_1$, so $\mathcal{M}$ is finite and S is a finite union of arithmetic progressions from a_1 on. Periodicity (from $n = 1!$) follows at once.

Current best

Complete proof. With $\mathcal{M}$ the (finite, as proved) set of minimal transversals of $\{P(s) : s \in S\}$, one may take $L = M := \prod_{p \leq a_1, p \text{ prime}} p$ and $T = \#\{r \in \{0, \dots, M-1\} : \exists Q \in \mathcal{M}, d_Q \mid r\} (\geq 1)$; then $a_{n+T} = a_n + L$ for **every** $n \geq 1$. Verified numerically for ~ 70 seeds (all $2 \leq a_1 \leq 60$ and assorted larger ones, ranges up to $4 \cdot 10^5$).

Full proof

Throughout, “prime set” means a finite set of prime numbers. For an integer $n > 1$ let $P(n)$ denote the set of primes dividing n ; it is finite and nonempty. For a prime set X put

$$\pi(X) := \prod_{p \in X} p \quad (\pi(\emptyset) = 1).$$

Note that for an integer $m \geq 1$ and a prime set X ,

$$X \subseteq P(m) \iff \pi(X) \mid m, \quad (0)$$

since $\pi(X)$ is a product of distinct primes.

By its definition the sequence is strictly increasing, so all terms are distinct and $a_n \rightarrow \infty$. Let

$$S := \{a_1, a_2, a_3, \dots\}.$$

Every element of S is $\geq a_1 \geq 2$.

Step 1: Basic properties of S

Lemma 1 (pairwise non-coprimality). For all $s, s' \in S$ we have $\gcd(s, s') > 1$.

Proof. If $s = s'$ then $\gcd(s, s') = s > 1$. If $s = a_i, s' = a_j$ with $i < j$, then $j \geq 2$ and, by the defining property of a_j (applied with $n = j - 1$), $\gcd(a_j, a_i) > 1$. $\square$

Lemma 2 (completeness). $S = \{m \in \mathbb{Z} : m \geq a_1 \text{ and } \gcd(m, s) > 1 \text{ for all } s \in S\}$.

Proof. “ $\subseteq$ ”: every element of S is $\geq a_1$, and Lemma 1 applies.

“ $\supseteq$ ”: Let $m \geq a_1$ satisfy $\gcd(m, s) > 1$ for all $s \in S$, and suppose $m \notin S$. Then $m > a_1$. Since $a_n \rightarrow \infty$ and $a_1 < m$, the index

$$n := \max\{i \geq 1 : a_i < m\}$$

is well defined. By maximality of n we have $a_{n+1} \geq m$, and $a_{n+1} \neq m$ because $m \notin S$; hence $a_{n+1} > m$. On the other hand $m > a_n$ and $\gcd(m, a_i) > 1$ for all $i \leq n$ (these a_i lie in S), so m is an integer greater than a_n having a nontrivial common factor with each of $a_1, \dots, a_n$. By the minimality in the definition of a_{n+1} , $a_{n+1} \leq m$ — a contradiction. $\square$

Lemma 3 (downward witnesses; this is where greediness enters). If $m > a_1$ and $m \notin S$, then there exists $s \in S$ with $s < m$ and $\gcd(s, m) = 1$.

Proof. Define $n := \max\{i \geq 1 : a_i < m\}$ as in Lemma 2; as there, $a_n < m < a_{n+1}$. If we had $\gcd(m, a_i) > 1$ for every $i \leq n$, then m would be a candidate in the definition of a_{n+1} , forcing $a_{n+1} \leq m$, a contradiction. Hence $\gcd(m, a_i) = 1$ for some $i \leq n$, and $s := a_i \leq a_n < m$ works. $\square$

Step 2: Transversals and the structure of S

Definition. A prime set X is a *transversal* if $X \cap P(s) \neq \emptyset$ for every $s \in S$. A *member* is a transversal Q such that no proper subset of Q is a transversal. Let $\mathcal{M}$ denote the set of all members, and for $Q \in \mathcal{M}$ put $d_Q := \pi(Q)$.

Two immediate remarks:

- $\emptyset$ is not a transversal (as $S \neq \emptyset$ and each $P(s)$ is disjoint from $\emptyset$); hence every member is nonempty, so $d_Q \geq 2$.
- **Every transversal X contains a member.** Indeed, among the finitely many subsets of X that are transversals (a nonempty collection, containing X), choose one, Q , that is inclusion-minimal in this collection. If some proper subset $Q'' \subsetneq Q$ were a transversal, then $Q'' \subseteq X$ would contradict the minimal choice of Q . So Q is a member. (*)

Lemma 4 (exact description of S). $\mathcal{M} \neq \emptyset$, and for every integer $m \geq a_1$:

$$m \in S \iff d_Q \mid m \text{ for some } Q \in \mathcal{M}.$$

Proof. “ $\Leftarrow$ ”: Suppose $d_Q \mid m$, i.e. (by (0)) $Q \subseteq P(m)$. For every $s \in S$, since Q is a transversal, $\emptyset \neq Q \cap P(s) \subseteq P(m) \cap P(s)$, i.e. $\gcd(m, s) > 1$. As $m \geq a_1$, Lemma 2 gives $m \in S$.

“ $\Rightarrow$ ”: Let $m \in S$. By Lemma 1, $\gcd(m, s) > 1$ for every $s \in S$, i.e. $P(m)$ is a transversal. By (*) it contains a member Q , and $Q \subseteq P(m)$ means $d_Q \mid m$ by (0).

Finally, $P(a_1)$ is a transversal (Lemma 1), so it contains a member; thus $\mathcal{M} \neq \emptyset$. $\square$

Lemma 5 (members pairwise intersect). For all $Q, Q' \in \mathcal{M}$ we have $Q \cap Q' \neq \emptyset$.

Proof. Choose distinct primes $p, p' \notin Q \cup Q'$ (possible: there are infinitely many primes). Choose integers $e, e' \geq 1$ with $m := d_Q p^e \geq a_1$ and $m' := d_{Q'} p'^{e'} \geq a_1$. By Lemma 4 (“ $\Leftarrow$ ”), $m, m' \in S$. By Lemma 1, $\gcd(m, m') > 1$, so some prime u lies in

$$P(m) \cap P(m') = (Q \cup \{p\}) \cap (Q' \cup \{p'\}).$$

Since $p \neq p'$, $p \notin Q'$ and $p' \notin Q$, necessarily $u \in Q \cap Q'$. $\square$

Step 3: The key lemma — cheap witnesses

Lemma 6 (cheap witness). Let $Q \in \mathcal{M}$ with $|Q| \geq 2$, and let $t \in Q$. Then there exists $Q' \in \mathcal{M}$ with

$$Q' \cap Q = \{t\} \quad \text{and} \quad d_{Q'} < \frac{a_1 d_Q}{t}.$$

Proof. Put $X := Q \setminus \{t\}$; then $X \neq \emptyset$ and $\pi(X) = d_Q/t \geq 2$.

First, **no member is a subset of X** : if $M \in \mathcal{M}$ and $M \subseteq X \subsetneq Q$, then M would be a transversal that is a proper subset of the member Q , contradicting the definition of a member.

Let $e \geq 1$ be minimal with $m := \pi(X)^e > a_1$. Then

$$m \leq a_1 \pi(X) : \quad (1)$$

if $e = 1$ this is clear since $m = \pi(X) \leq a_1 \pi(X)$; if $e \geq 2$, minimality of e gives $\pi(X)^{e-1} \leq a_1$, so $m = \pi(X)^{e-1} \cdot \pi(X) \leq a_1 \pi(X)$.

Since $P(m) = X$ contains no member and $m > a_1$, Lemma 4 gives $m \notin S$. By Lemma 3 there exists $s \in S$ with $s < m$ and $\gcd(s, m) = 1$, i.e.

$$P(s) \cap X = \emptyset. \quad (2)$$

By Lemma 4 (“ $\Rightarrow$ ”) there is a member $Q' \subseteq P(s)$. Then by (2), $Q' \cap X = \emptyset$, and by (0), $d_{Q'} \mid s$, hence using (1):

$$d_{Q'} \leq s < m \leq a_1 \pi(X) = \frac{a_1 d_Q}{t}.$$

By Lemma 5, $Q' \cap Q \neq \emptyset$; since $Q = X \cup \{t\}$ and $Q' \cap X = \emptyset$, this forces $Q' \cap Q = \{t\}$. $\square$

Step 4: Members contain no prime exceeding a_1

Lemma 7. Every prime belonging to some member is $\leq a_1$.

Proof. **Case A: some member is a singleton $\{p\}$.** Let $M \in \mathcal{M}$ be arbitrary. By Lemma 5, $M \cap \{p\} \neq \emptyset$, i.e. $p \in M$. Then $\{p\} \subseteq M$ is a transversal, and since no proper subset of the member M is a transversal, $M = \{p\}$. Hence $\mathcal{M} = \{\{p\}\}$. Applying Lemma 4 (“ $\Rightarrow$ ”) to $a_1 \in S$ gives $p \mid a_1$, so $p \leq a_1$, and the lemma holds in this case.

Case B: every member has at least two elements. Suppose, for contradiction, that some member Q_0 contains a prime $q > a_1$. Define recursively, for $i \geq 0$: $Q_{i+1} :=$ the member provided by Lemma 6 applied to $Q := Q_i$ and $t := q$. This is legitimate by induction: $q \in Q_i$ (for $i = 0$ by assumption; for $i \geq 1$ because $Q_i \cap Q_{i-1} = \{q\}$), and $|Q_i| \geq 2$ by the Case B hypothesis. Lemma 6 yields

$$d_{Q_{i+1}} < \frac{a_1}{q} d_{Q_i} \quad (i \geq 0), \quad \text{hence} \quad d_{Q_i} < \left(\frac{a_1}{q}\right)^i d_{Q_0} \quad (i \geq 1).$$

Since $q > a_1$, we have $a_1/q < 1$, so $d_{Q_i} \rightarrow 0$ as $i \rightarrow \infty$. But $q \in Q_i$ implies $d_{Q_i} \geq q \geq 2$ for all i — a contradiction. $\square$

Lemma 8 (finiteness). $\mathcal{M}$ is finite and nonempty. Moreover, with

$$M := \prod_{\substack{p \text{ prime} \\ p \leq a_1}} p \quad (\geq 2, \text{ since } a_1 \geq 2),$$

every $Q \in \mathcal{M}$ satisfies $d_Q \mid M$.

Proof. By Lemma 7, every member is a subset of the finite set $\Pi := \{p \text{ prime} : p \leq a_1\}$; hence $\mathcal{M} \subseteq 2^\Pi$ is finite. It is nonempty by Lemma 4. Each d_Q is a product of distinct primes $\leq a_1$, hence divides M . $\square$

Step 5: Conclusion — exact linear periodicity

Define

$$R := \{r \in \{0, 1, \dots, M-1\} : d_Q \mid r \text{ for some } Q \in \mathcal{M}\}, \quad T := |R|, \quad L := M.$$

Since $d_Q \mid 0$ for any $Q \in \mathcal{M} \neq \emptyset$, we have $0 \in R$, so $T \geq 1$; and $L = M \geq 2$. Thus T, L are positive integers.

Because $d_Q \mid M$ for all $Q \in \mathcal{M}$ (Lemma 8), for any integer m we have $d_Q \mid m \iff d_Q \mid (m \bmod M)$. Hence Lemma 4 can be restated: for every integer $m \geq a_1$,

$$m \in S \iff (m \bmod M) \in R. \quad (3)$$

In particular,

$$\text{for every integer } m \geq a_1 : \quad m \in S \iff m + M \in S. \quad (4)$$

For $k \geq 0$ let $I_k := \{a_1 + kM, a_1 + kM + 1, \dots, a_1 + (k+1)M - 1\}$, a block of M consecutive integers, all $\geq a_1$. Each I_k contains exactly one integer from each residue class modulo M , so by (3),

$$|S \cap I_k| = |R| = T \quad \text{for every } k \geq 0. \quad (5)$$

Also $S \subseteq \bigcup_{k \geq 0} I_k$, because $\min S = a_1$.

Claim. For every $k \geq 0$, the elements of S smaller than $a_1 + kM$ are exactly $a_1, a_2, \dots, a_{kT}$ (so $S \cap I_k = \{a_{kT+1} < a_{kT+2} < \dots < a_{(k+1)T}\}$).

Proof of Claim. Induction on k . For $k = 0$ the statement is vacuously true (S has no elements below a_1). Assume it for k . The elements of S smaller than $a_1 + (k+1)M$ are those smaller than $a_1 + kM$ — namely $a_1, \dots, a_{kT}$ — together with $S \cap I_k$, which by (5) has exactly T elements, each larger than every element of S below $a_1 + kM$. Since $a_1 < a_2 < \dots$ enumerates S in increasing order, $S \cap I_k = \{a_{kT+1}, \dots, a_{(k+1)T}\}$ and the elements below $a_1 + (k+1)M$ are $a_1, \dots, a_{(k+1)T}$. $\square$

By (4), the map $\varphi(m) := m + M$ maps $S \cap I_k$ into $S \cap I_{k+1}$ (if $m \in S \cap I_k$ then $m \geq a_1$, so $m + M \in S$, and clearly $m + M \in I_{k+1}$), and the map $m' \mapsto m' - M$ maps $S \cap I_{k+1}$ into $S \cap I_k$ (if $m' \in S \cap I_{k+1}$ then $m' - M \geq a_1$ and $m' - M \in S$ by (4) applied to $m := m' - M$). These maps are mutually inverse, so $\varphi : S \cap I_k \rightarrow S \cap I_{k+1}$ is a strictly increasing bijection between two T -element sets. A strictly increasing bijection between finite sets listed in increasing order matches the j -th smallest element to the j -th smallest element; combined with the Claim this gives

$$a_{(k+1)T+j} = a_{kT+j} + M \quad \text{for all } k \geq 0, 1 \leq j \leq T.$$

Finally, let $n \geq 1$ be arbitrary and write $n = kT + j$ with $k \geq 0$ and $1 \leq j \leq T$. Then

$$a_{n+T} = a_{(k+1)T+j} = a_{kT+j} + M = a_n + L.$$

Thus $a_{n+T} = a_n + L$ for every positive integer n , with the positive integers T and L defined above.

■

Remarks

- The proof shows more: $S = \{m \geq a_1 : d_Q \mid m \text{ for some member } Q\}$ is *exactly* a finite union of arithmetic progressions intersected with $[a_1, \infty)$ — the periodic description is valid from the very first term, which is why no “eventual” qualifier is needed.
- Greediness is used exactly once, in Lemma 3 (every excluded integer has a **smaller** coprime witness in S); this is essential: there exist “self-consistent” sets S (satisfying Lemma 2 but not Lemma 3) built from an infinite self-blocking clutter which are not unions of finitely many progressions. See `approaches/abstract-clutter-dead-end.md`.
- Numerical verification: `code/verify.py` confirms Lemmas 4, 5, 7 and the Theorem (with these exact T, L up to replacing M by $\text{lcm}_Q d_Q$, a divisor of M ; both work) for all $2 \leq a_1 \leq 60$ and for $a_1 \in \{77, 91, 101, 105, 121, 125, 143, 169, 255, 385, 663, 1001, 1155\}$, over ranges up to $4 \cdot 10^5$.